

Post-Quantum Cryptography

Preparing for the Quantum Era in Secure Communications

Randy Balzer

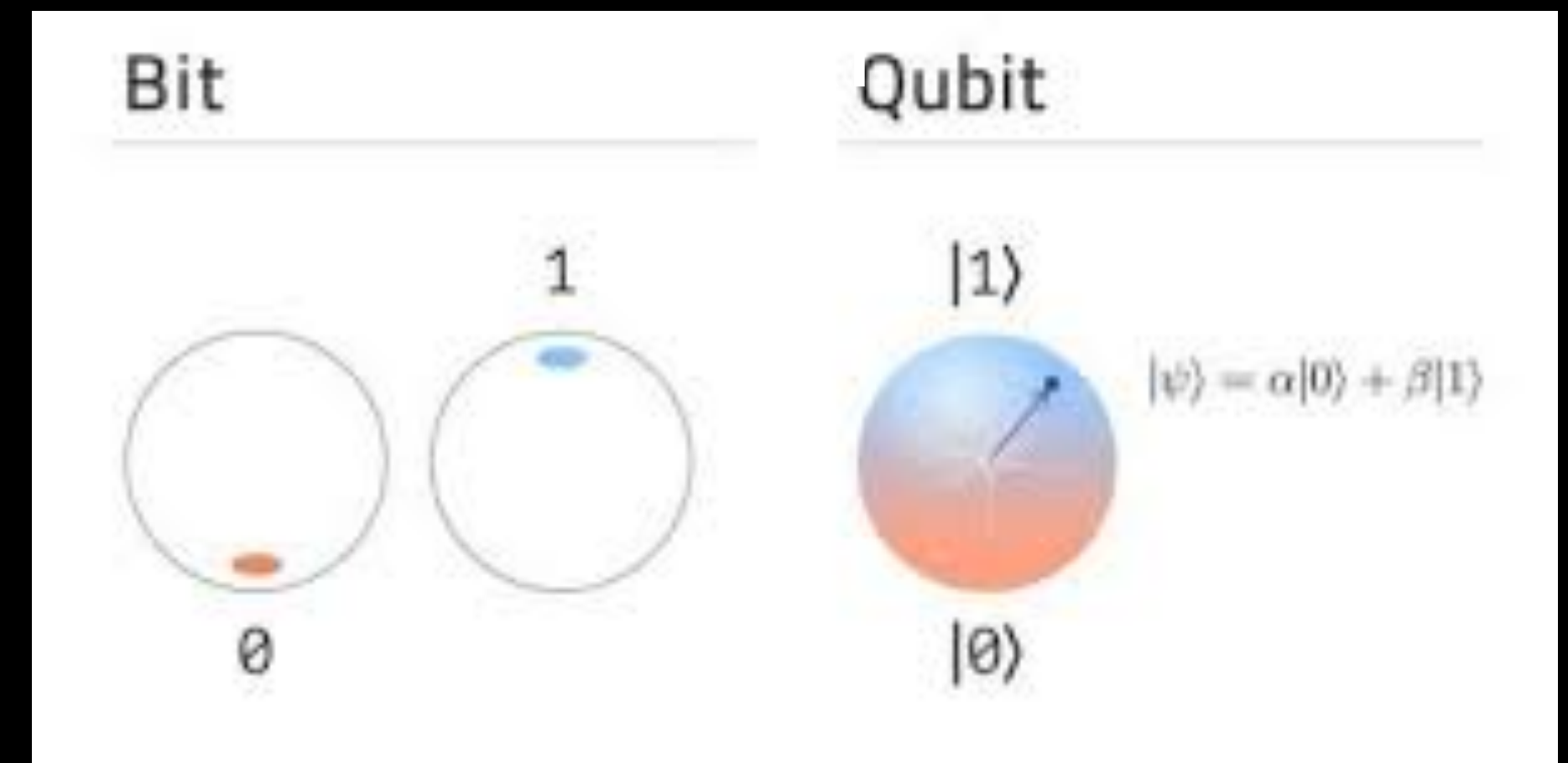
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Overview of Classical Cryptography

- Classical cryptography relies on computational hardness assumptions (e.g., integer factorization, discrete logarithms).
- Key systems: RSA (public-key encryption), ECC (elliptic curve cryptography), AES (symmetric encryption).
- These ensure confidentiality, integrity, and authentication in digital communications.

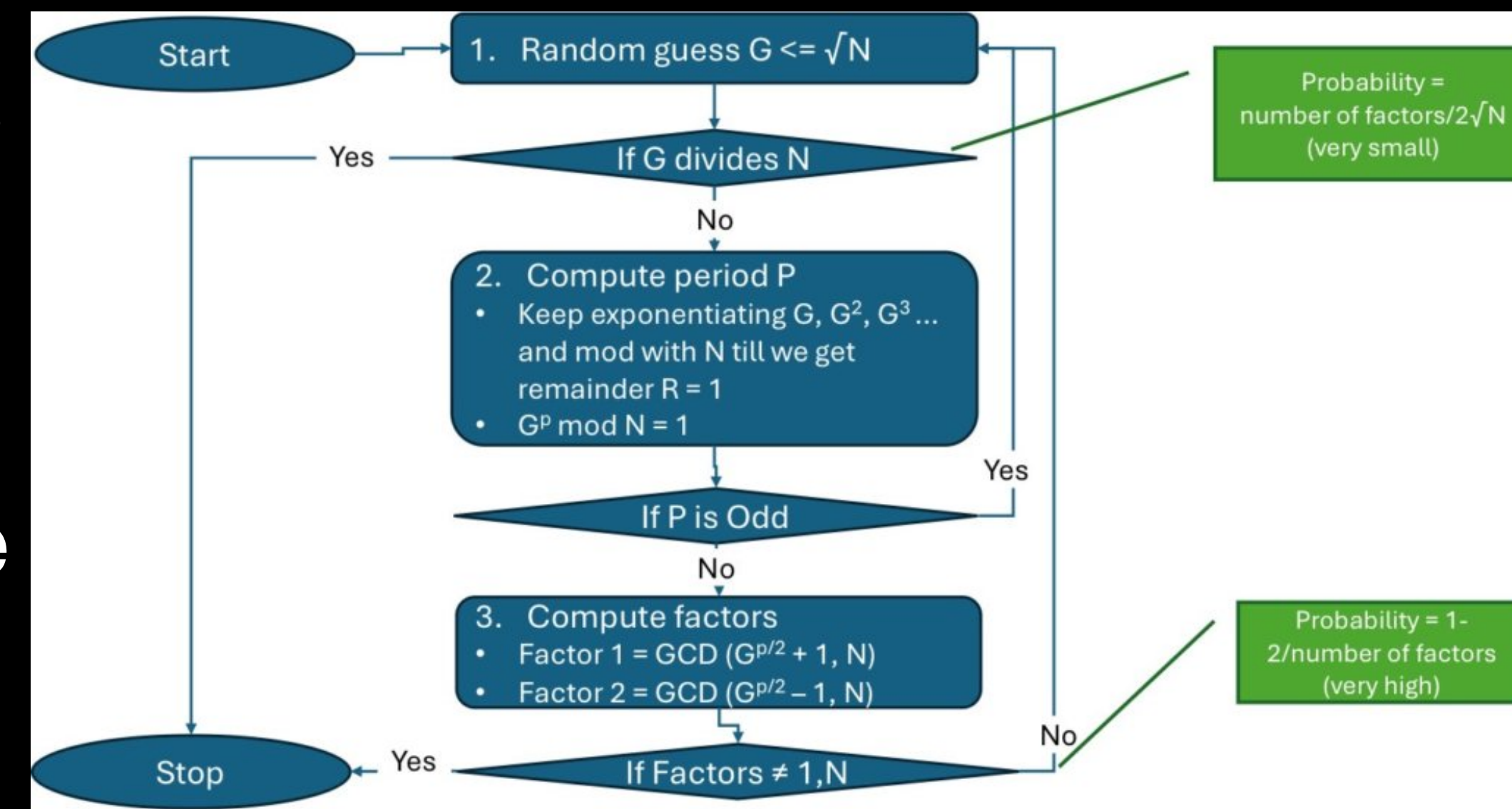
Introduction to Quantum Computing

- Quantum computers use qubits, superposition, and entanglement for parallel processing.
- Potential to solve certain problems exponentially faster than classical computers.
- Examples: Optimization, simulation, and cryptography-breaking algorithms.



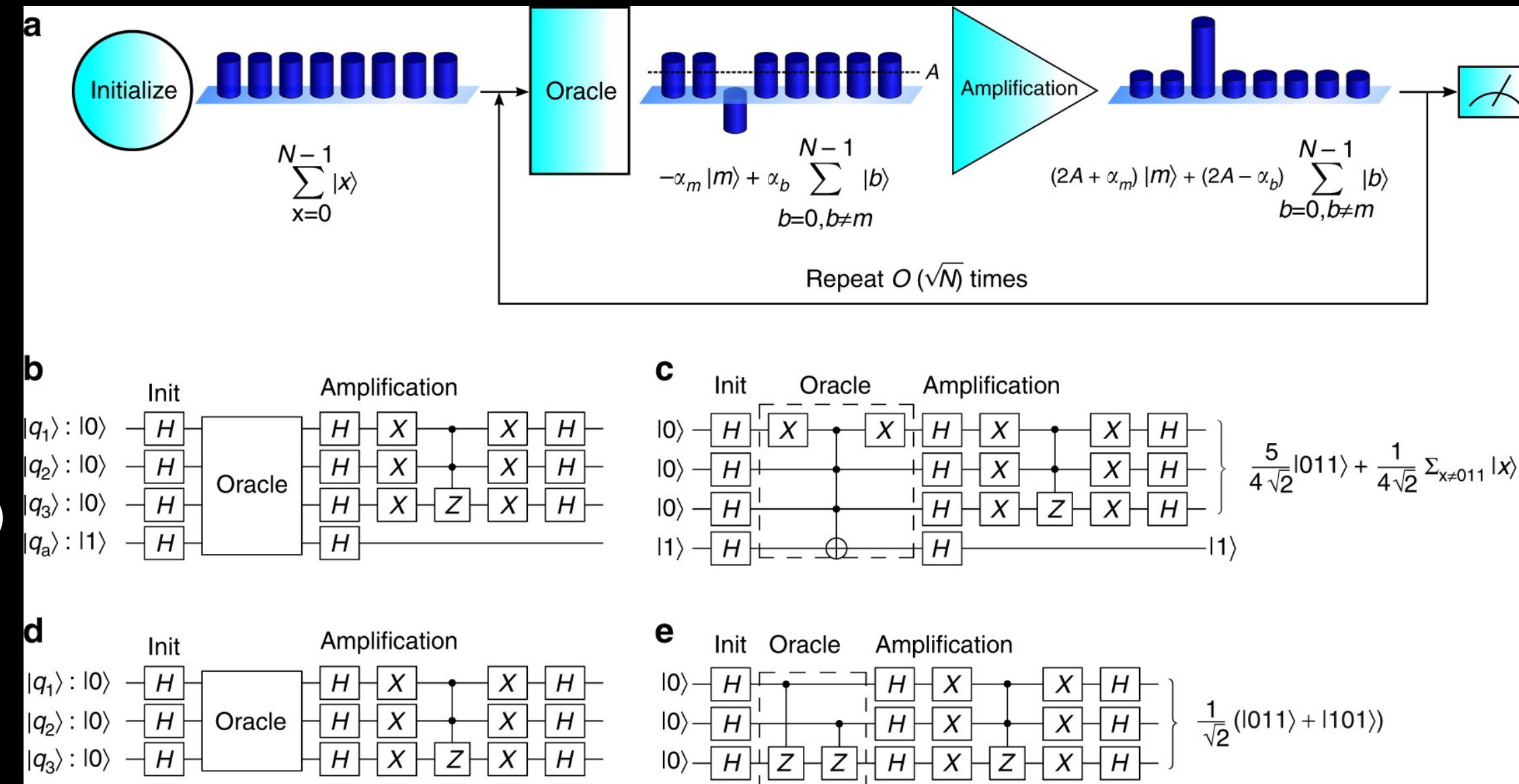
Quantum Threats: Shor's Algorithm

- Developed by Peter Shor in 1994.
- Efficiently factors large integers and solves discrete logarithm problems on quantum computers.
- Breaks RSA and ECC with polynomial time complexity.
- Impact: Renders public-key cryptography insecure at scale.



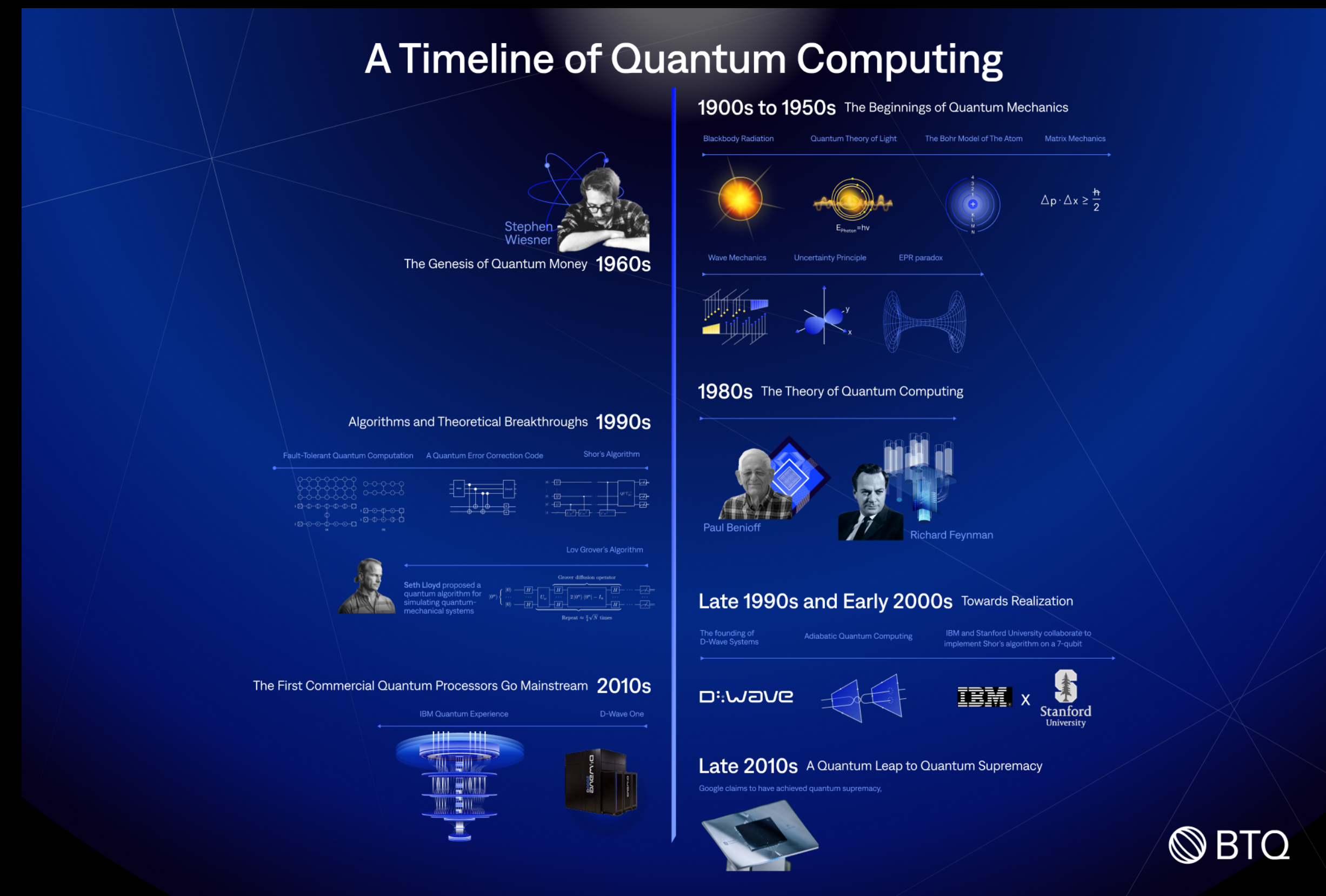
Quantum Threats: Grover's Algorithm

- Developed by Lov Grover in 1996.
- Provides quadratic speedup for unstructured search problems.
- Affects symmetric cryptography (e.g., halves effective key length of AES-256 to 128 bits).
- Mitigation: Increase key sizes for symmetric algorithms.



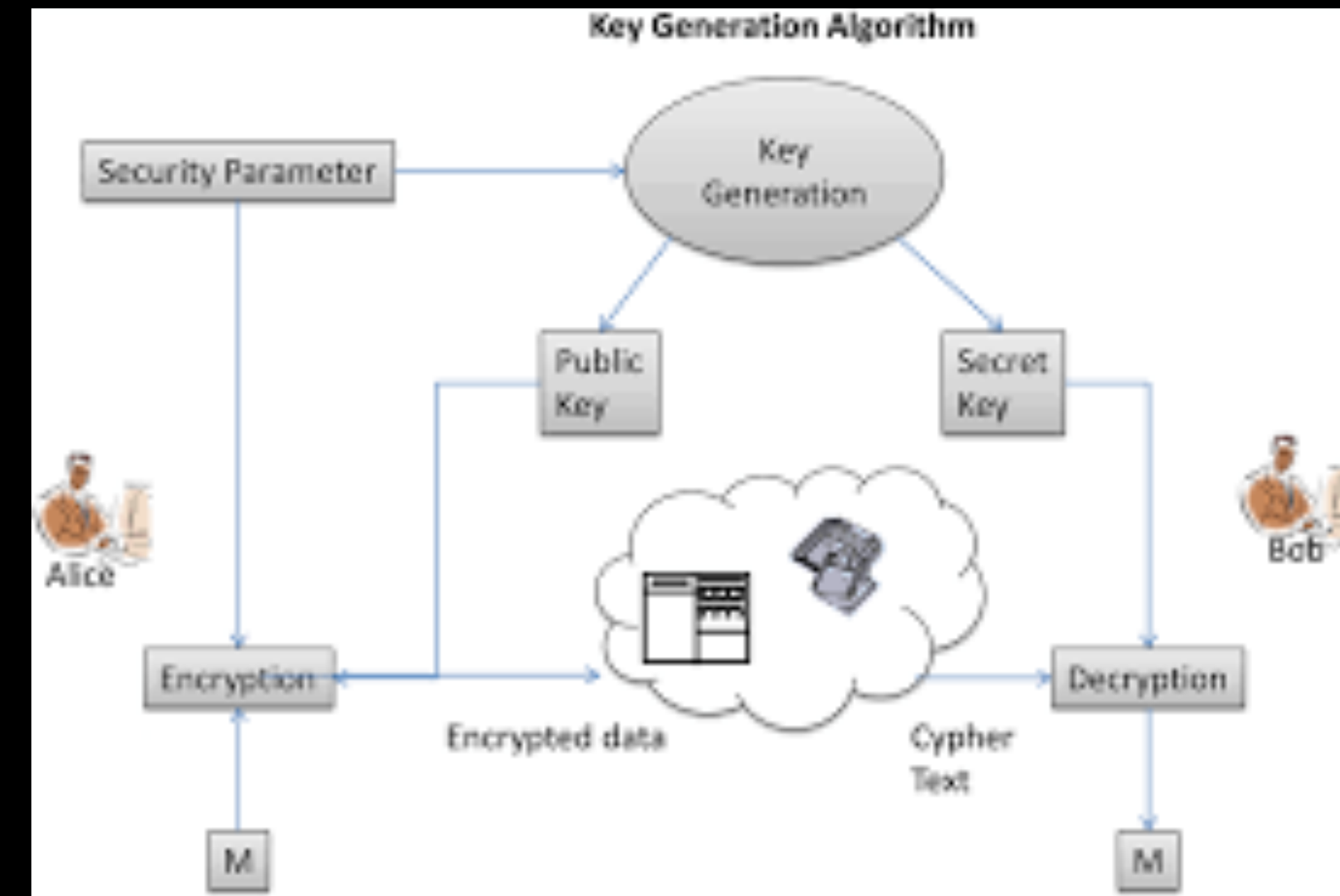
The Need for Post-Quantum Cryptography (PQC)

- PQC: Cryptographic primitives resistant to both classical and quantum attacks.
- Goals: Maintain security in a post-quantum world without quantum hardware.
- Timeline: Quantum computers may become viable in 10-20 years; preparation must start now.



Categories of PQC Algorithms

- Lattice-based: Based on shortest vector problems (e.g., NTRU, Kyber).
- Hash-based: Use one-way hash functions (e.g., SPHINCS+).
- Code-based: Rely on error-correcting codes (e.g., McEliece).
- Multivariate: Solve systems of multivariate equations (e.g., Rainbow).
- Isogeny-based: Use elliptic curve isogenies (e.g., SIDH, but with known vulnerabilities).

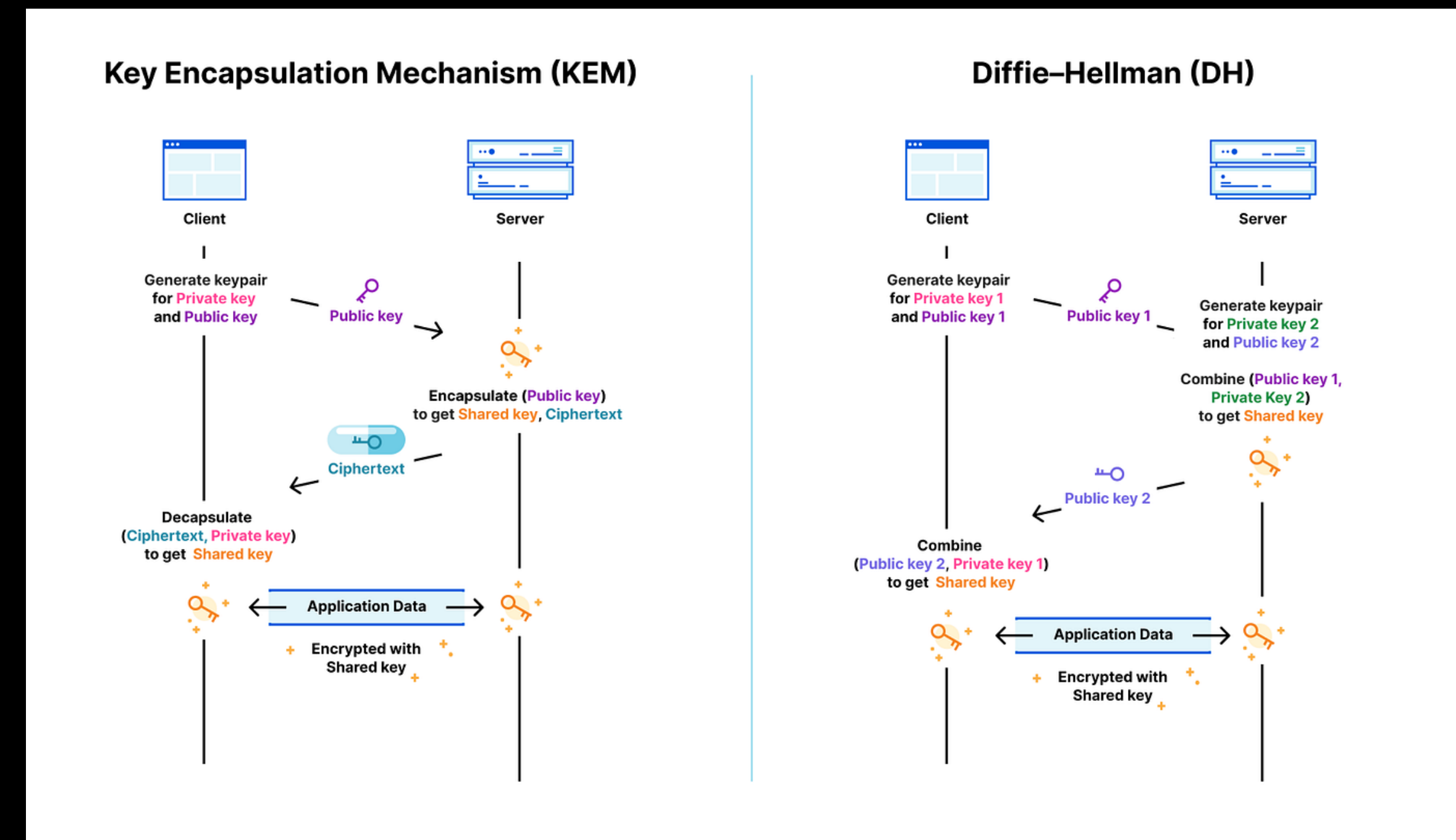


NIST Standardization Process

- NIST (National Institute of Standards and Technology) leads global PQC standardization.
- Rounds: Started in 2016; Round 4 ongoing as of 2025.
- Selected standards: CRYSTALS-Kyber (encryption), CRYSTALS-Dilithium (signatures), FALCON (signatures), SPHINCS+ (signatures).
- Process: Public submissions, cryptanalysis, and performance evaluations.

Example: CRYSTALS-Kyber

- Lattice-based key encapsulation mechanism (KEM).
- Security: Based on module learning with errors (MLWE) problem.
- Advantages: Efficient key exchange, small key sizes.
- Use cases: Hybrid with classical schemes for transition.

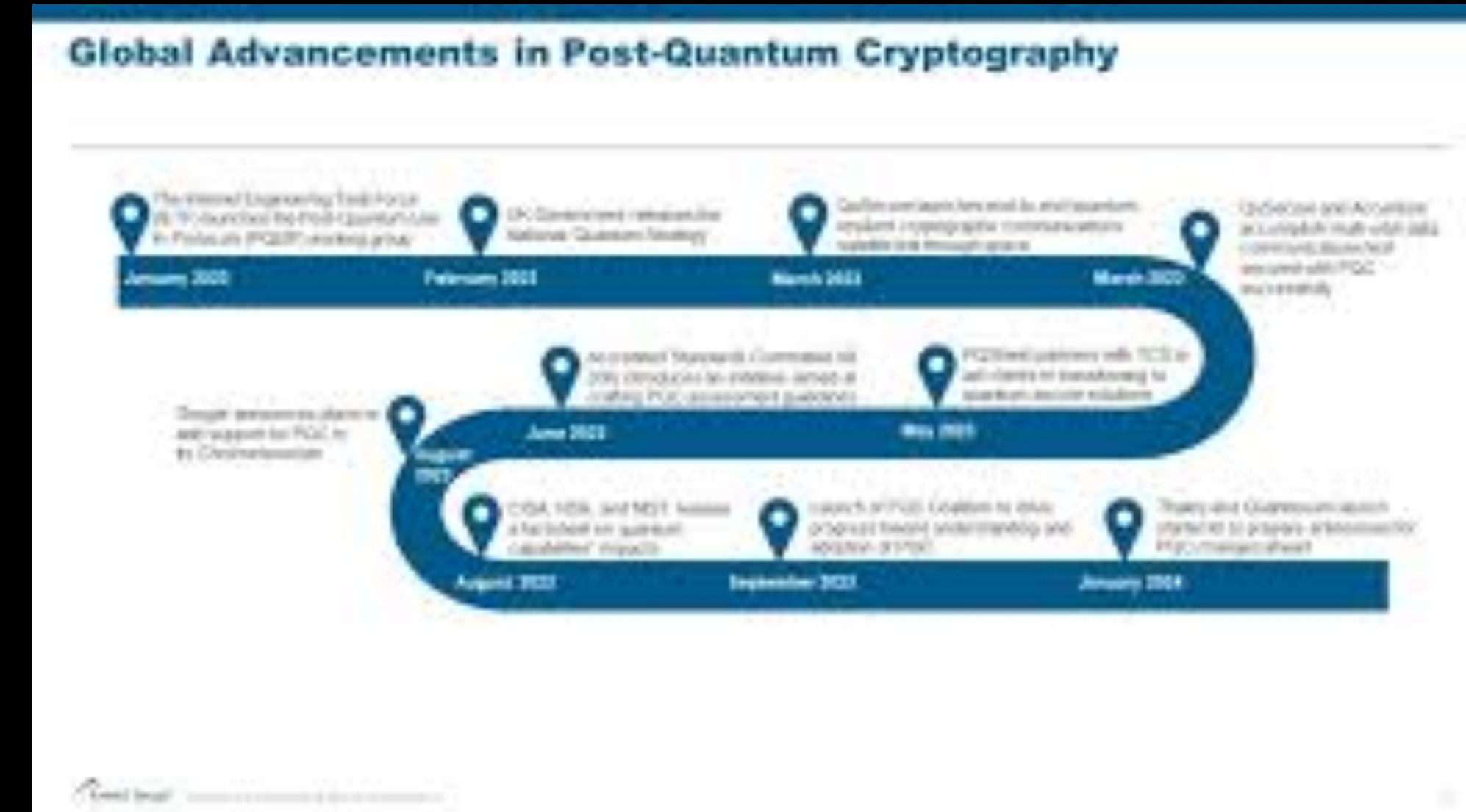


Challenges in PQC Adoption

- Performance overhead: Larger keys and signatures in some schemes.
- Side-channel attacks: Quantum resistance does not imply classical security.
- Integration: Updating legacy systems and protocols.
- Cryptanalysis: Ongoing research to verify long-term security.

Future Directions and Research

- Quantum key distribution (QKD): Complementary to PQC, uses quantum mechanics for key exchange.
- Hybrid cryptography: Combine classical and PQC for backward compatibility.
- Open areas: Fully homomorphic encryption, zero-knowledge proofs in quantum settings.



Conclusion and Q&A

- Summary: Quantum computing threatens classical crypto; PQC provides resilient alternatives.
- Key takeaway: Proactive adoption is crucial for long-term security.
- Resources: NIST PQC website, research papers on ePrint.
- Questions?